

Text as Data for Economic Research

From Corpus to Validated Measure

Class 1. Counting, Dictionaries, and Validation
June 5, 2026

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Plan for Today

- Logistics and course overview
- Why text as data: core tasks and key principles
- Selecting and representing text
- Bag of Words
- Principles of Discovery: discriminating words
- Principles of Measurement: dictionaries, sentiment, validation

Organization

Logistics and Materials

- **Course GitHub folder** — slides, notebooks, problem sets, reading list
→ github.com/tizianorotesi/text-as-data (link TBA)
- **Google Colab** — one notebook per class, runs in the browser.
No local install required.
- **Four problem sets** — applied, **ungraded, not mandatory**. A way to practice the methods and adapt them to your own work.

No exam, no research proposal — this is a short, hands-on module.

Readings

Textbooks — useful if you want to go deeper, none is required:

- Grimmer, J., Roberts, M. E., & Stewart, B. M. (2022). *Text as Data: A New Framework for Machine Learning and the Social Sciences*.
 - Main social-science reference; the principles in this course follow it.
- Jurafsky, D., & Martin, J. H. (2023). *Speech and Language Processing*, 3rd ed. (available online).
 - More advanced NLP depth; useful for Classes 3–4.

Papers are listed per session in the syllabus and on GitHub.

Why Text as Data

Text Is Now an Economic Variable

- Earnings calls, FOMC minutes, legislative records, newspaper archives, contracts, reviews — all are data.
- Text lets us *measure* things we previously could not: sentiment, uncertainty, attention, slant, narratives.
- The hard part is not running the model — it is producing a measure that is **reliable and replicable** enough for credible inference.

Hassan et al. (2025); Gentzkow, Kelly & Taddy (2019).

The Four Computational Tasks

1. **Concept detection** — is a concept present? (sentiment, uncertainty)
2. **Document similarity** — how alike are two texts?
3. **How concepts relate** — co-occurrence, association, analogy
4. **Text** $\leftrightarrow$ **metadata** — link text to authors, dates, outcomes

Representations (counts $\rightarrow$ ML $\rightarrow$ embeddings $\rightarrow$ LLMs) are tools in service of these tasks. We start with the most transparent.

Ash and Hansen (2023).

Key Principles for Text Analysis

From Grimmer, Roberts, and Stewart (2022):

1. *Social science theories and substantive knowledge* are essential for research design
2. Text analysis *does not replace humans* — it augments them
3. Building, refining, and testing theories requires *iteration and cumulation*
4. Text analysis methods distill *generalizations from language*
5. *The best method depends on the task*
6. *Validations* are essential and depend on the theory and the task

Selecting and Representing Text

Building a Corpus

- A **corpus** is a structured collection of documents, used for text analysis and language modeling
- Each **document** in the corpus — such as a newspaper article, earnings call, or legislative speech — is a distinct unit of observation

Key decisions:

- The specific population reflected in the corpus
- What to pay attention to and the risks

Population and Quantities of Interest

Population of interest

+

Research question



Corpus

+

Quantity of interest



Question-Specific Corpus Construction

The usefulness of a corpus depends on the question the researcher wants to answer and the population they want to study.

For a given research question:

1. What would be the ideal population (of text) to study?
2. How does the corpus I can build differ from this population?

For a given corpus:

1. Determine the specific population reflected in the corpus
2. What is interesting about the population I can study with it?

No Values-Free Corpus Construction

There is no value-free construction of a corpus. Selecting which documents to include has ramifications.

- The text reflects the process that produced it — and its biases
- Language may reflect the biases of its authors
- **Avoiding sample selection** requires understanding how texts enter the corpus

Avoiding Sample Selection

These considerations are important to avoid introducing selection bias into our analyses:

- Resource Bias
- Incentive Bias
- Medium Bias
- Retrieval Bias

Resource Bias

Text often reflects the perspectives of populations with more resources to produce, record, and store documents.

Some examples:

- **Historical Document Preservation:** Requires resources for storage and maintenance
- **Data Creation Costs:** Expenses in transcribing meetings, scanning printed documents, digitizing archives...

E.g. *Chronicling America / American Stories*: only newspapers in the public domain, with imperfect OCR.

Incentive Bias

Strategic behavior influences the creation and public availability of documents.

→ It's important to recognize and understand these incentives when analyzing data

Examples:

- **Social Media:** users share content to portray happy moments or successful self-images
- **Earnings Reports:** e.g. the trend of labeling every company an AI company, regardless of actual focus
- **Electoral Campaigns:** messaging is typically highly strategic and carefully planned

Medium Bias

The Medium is the Message
- Marshall McLuhan

The technologies and types of mediums in which texts are recorded influence the nature of the content produced.

Examples:

- **Social Media:**

- Short messages: hashtags, differences across languages
- Focus on eliciting immediate reactions from readers

- **Radio vs. Podcasts:**

- Radio is often faster-paced with shorter discussions

Retrieval Bias

The method used to retrieve data can significantly influence the selection of a corpus and potentially introduce bias.

Examples:

- Documents retrieved using **keyword search**
 - What if certain keywords are systematically excluded from specific types of documents?
- **Social media:** how to select users?
 - Following popular accounts → bias towards certain interests

No Right Way to Represent Text

There is no one right way to represent text for all research questions.

- What specific concept are you aiming to capture?
- What is the most straightforward method to achieve this?
 - Frequency of a single word?
 - Compare usage of selected groups of words?
 - Study the context in which some words are used?

Note: *You will always need to throw away some information.*

Representation is modeling.

Bag of Words

Documents as Bag of Words

- How can we extract the information we need from a collection of documents?
 - To analyze text data, we need to convert documents into a numerical format.
- The *Bag of Words* model is the most common way to represent text
- **Core Idea:** Count how many times each word appears in the document

Bag of Words Model Example

Take the following verses, consider them as your documents:

1. *How many roads must a man walk down*
2. *Before you call him a man?*
3. *How many seas must a white dove sail*
4. *Before she sleeps in the sand?*

	How	many	roads	must	a	man	walk	down	Before	you	call	him	seas	white	dove	sail	she	sleeps	in	the	sand
Doc1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
Doc2	0	0	0	0	1	1	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0
Doc3	1	1	0	1	1	0	0	0	0	0	0	0	1	1	1	1	0	0	0	0	0
Doc4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	1	1	1	1

Document-Feature Matrix

	How	many	roads	must	a	man	walk	down	Before	you	call	him	seas	white	dove	sail	she	sleeps	in	the	sand
Doc1	1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0
Doc2	0	0	0	0	1	1	0	0	1	1	1	1	0	0	0	0	0	0	0	0	0
Doc3	1	1	0	1	1	0	0	0	0	0	0	0	1	1	1	1	0	0	0	0	0
Doc4	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	1	1	1	1	1

This is an example of a document-feature matrix:

- Rows $\rightarrow$ Documents
- Columns $\rightarrow$ Features — in this case words

Building a Document-Feature Matrix

Considerations for Constructing a Document-Feature Matrix:

These choices critically influence the outcomes of subsequent analyses.

- **Unit of Analysis:** What defines a “document”
 - **Tokenization:** Identifying basic units in documents
 - **Complexity Reduction:**
 - Standardize text (lowercase, remove punctuation, stop words)
 - Lemmatization/Stemming and frequency-based filtering
- ! The specific approach should be guided by your research question

Unit of Analysis

What should be the **unit of analysis**?

- Documents can be articles, books, papers, speeches, earnings calls, tweets...
 - Availability may vary
 - The choice may not always align with your research objectives
- This choice affects computational costs and model efficiency

Quick Question: Can you think of scenarios where smaller/larger units might be more appropriate?

Tokens: Breaking Up a Document

How should we segment a document?

- **Words:**
 - In English: typically separated by whitespace
 - In other languages: may follow different segmentation rules
- **Syllables / word pieces:**
 - Often used in more advanced models like FastText, GPT, or BERT
 - We will not consider this for now
- **N-Grams** (ordered sets of n words): e.g., `interest_rate`, `war_on_terror`
 - Can include all possible combinations → the set of features may become too large!
 - Need criteria to choose which n-grams to include

Key Principle: Retain essential information and simplify where possible to manage complexity.

1. **Lowercase**
2. **Remove Punctuation and Numbers**
3. **Remove Stop Words**
4. **Lemmatization/Stemming**
5. **Filter by Frequency**

Key Principle: Retain essential information and simplify where possible to manage complexity.

1. **Lowercase**

- Usually, consider capitalized and non-capitalized versions of a word as equivalent
- May not always be the case: “march” vs. “March”

2. **Remove Punctuation and Numbers**

3. **Remove Stop Words**

4. **Lemmatization/Stemming**

5. **Filter by Frequency**

Key Principle: Retain essential information and simplify where possible to manage complexity.

1. **Lowercase**
2. **Remove Punctuation and Numbers**
 - These elements may matter in very specific cases only
3. **Remove Stop Words**
4. **Lemmatization/Stemming**
5. **Filter by Frequency**

Key Principle: Retain essential information and simplify where possible to manage complexity.

1. **Lowercase**
2. **Remove Punctuation and Numbers**
3. **Remove Stop Words**
 - E.g. and, the, ...
 - These words may be important, possibly in combination with other words
4. **Lemmatization/Stemming**
5. **Filter by Frequency**

Key Principle: Retain essential information and simplify where possible to manage complexity.

1. **Lowercase**
2. **Remove Punctuation and Numbers**
3. **Remove Stop Words**
4. **Lemmatization/Stemming**
 - Stemming: keep only the initial part of the words according to simple rules
 - e.g. family and families → famili
 - cannot capture irregular verbs (saw, see) or the same form with different meanings (writing)
 - Lemmatization: canonical form of a word (as in a dictionary)
5. **Filter by Frequency**

Key Principle: Retain essential information and simplify where possible to manage complexity.

1. **Lowercase**
2. **Remove Punctuation and Numbers**
3. **Remove Stop Words**
4. **Lemmatization/Stemming**
5. **Filter by Frequency**
 - Remove rare/common words
 - Important choice: trade-off between computational efficiency and informativeness

After Preprocessing



	how	mani	road	must	man	walk	down	befor	call	sea	white	dove	sail	sleep	sand
Doc1	1	1	1	1	1	1	1	0	0	0	0	0	0	0	0
Doc2	0	0	0	0	1	0	0	1	1	0	0	0	0	0	0
Doc3	1	1	0	1	0	0	0	0	0	1	1	1	1	0	0
Doc4	0	0	0	0	0	0	0	1	0	0	0	0	0	1	1

Stop words (a, you, him, she, in, the) removed. Here we **stem** (mani, famili, befor); in the notebook we **lemmatize** instead, which returns real dictionary words.

The Default Procedure Is Not Always Best

The approach above is not necessarily the best and may reduce performance:

- Order matters (e.g. stemming could be done before lowercasing)
- Stop words, capital letters, and rare words may be important
- **Variability in Methods:** No universal list of stop words or single best algorithm for stemming
- **Defining Frequency:** The criteria for “rare” and “common” words can be ambiguous
- **Handling Special Cases:** Consider unique elements like typos, emoticons, hashtags, and OCR errors in your data

Incorporating Linguistic Features

Incorporating linguistic knowledge into analysis is possible:

- **Sentence Detection:**

- Splits the document into sentences, which may serve as the basic units of observation instead of the entire document
- More challenging with paragraphs if they're not already formatted in the original document

- **Part of Speech (POS) Tags:**

- Key POS include verbs, nouns, pronouns, adjectives, adverbs, determiners, prepositions, and conjunctions
- The importance varies by task, e.g. using nouns for topics, adjectives for sentiment analysis

- **Dependency Parsing:**
 - Identifies grammatical relations between words: subject, object, modifier, and head
 - E.g. “Autonomous cars shift insurance liability toward manufacturers” → cars is subject of shift; liability is object of shift
 - Noun chunks, such as autonomous cars and insurance liability, are one useful feature derived from this structure
- **Named Entity Recognition (NER):** Recognizes specific entities, like “European Central Bank” or “U.K.”

! **spaCy** is a Python library that can do these operations

tf-idf Weighting

Raw counts over-reward ubiquitous words. Down-weight terms common across the corpus:

$$\text{tfidf}(t, d) = \underbrace{\text{tf}(t, d)}_{\text{freq. in doc}} \times \underbrace{\log \frac{N}{\text{df}(t)}}_{\text{inverse doc. freq.}}$$

- t = term; d = document; N = number of documents in the corpus
- $\text{tf}(t, d)$ = how often term t appears in document d
- $\text{df}(t)$ = number of documents containing term t

A document's *distinctive* vocabulary stands out — the standard input to similarity and to most ML models.

Principles of Discovery

Discovery

In the discovery phase, the main goal is to develop the research question.

- What do you want to measure?
- Which mechanisms are you interested in?

Examples of aspects to identify include:

- Ways of organizing the corpus → Clustering
- Common themes that link documents → Topic Models
- Concepts that characterize groups → Discriminating Words

Principles of Discovery

- **Context Relevance** — a discovery is good if it is useful for *your* problem
- **No Ground Truth** — there is no “correct” set of clusters or discriminating words
- **Judge the Concept, Not the Method**
 - There is no inherent value in using a specific method, apart from what it helps us learn about the conceptualization we are considering
- **Separate Data is Best**
 - Ideally, data used for discovery should be kept separate from data used with the goal of confirming these theories

Discriminating Words

Discover words that characterize the language use across groups of documents:

- Start from a known (or tentative) categorization: party, gender, newspaper, era, treatment status...
- Find which features characterize each group

This is the engine behind **media-slant** measurement.

Gentzkow & Shapiro (2010), “What Drives Media Slant?”

Distinctive vs. Prevalent

Key trade-off between how **distinctive** some language is vs. how **prevalent** it is:

! Risk of focusing on language that is used rarely but is unique to one group

tf-idf already surfaces distinctive vocabulary by down-weighting the ubiquitous — but raw frequency differences are dominated by noise.

We need to handle **estimation uncertainty** and trade off distinctiveness against frequency. Three methods, in order of increasing assumptions.

Mutual Information

The first measure of word separation that we will consider:

- Derived from **Information Theory**, it measures the association between two random variables
 - Intuition: the extent to which knowing one variable reduces uncertainty about the other
- In our context, **features** (words) and **categories** (groups) are treated as **random variables**

Mutual Information: Formula

- Let X and Y be two discrete random variables with distributions $P(X)$ and $P(Y)$ and joint distribution $P(X, Y)$.
- The Mutual Information $MI(X, Y)$ is defined as:

$$MI(X, Y) = \sum_{x \in X} \sum_{y \in Y} P(X=x, Y=y) \log_2 \left(\frac{P(X=x, Y=y)}{P(X=x) P(Y=y)} \right)$$

→ We never need the full sum. For discriminating words we use the **pointwise** version — a single word w and a single group c (next slide).

Example

- **Standard implementation:** look only at whether a feature appears in the document (0 or 1).
- Suppose we have the following document-feature matrix:

	family	nature	Jesus	health	...
Respondent 1	1	0	1	1	...
Respondent 2	1	0	1	1	...
Respondent 3	1	1	0	1	...
Respondent 4	1	0	0	0	...
Respondent 5	1	1	0	1	...
⋮	⋮	⋮	⋮	⋮	⋮

Example

Add a **categorization** — here, whether respondents are **conservatives or liberals**:

Group		family	nature	Jesus	health	...
Con →	Respondent 1	1	0	1	1	...
Con →	Respondent 2	1	0	1	1	...
Lib →	Respondent 3	1	1	0	1	...
Con →	Respondent 4	1	0	0	0	...
Lib →	Respondent 5	1	1	0	1	...
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Method 1 — PMI (fewest assumptions)

Pointwise Mutual Information focuses on *specific values* of the two variables: how much more a word w and a category c co-occur than under independence:

$$\text{PMI}(w, c) = \log_2 \frac{P(w, c)}{P(w) P(c)}$$

- In the example: *Con* and *Jesus* have a high PMI; *family* and *nature* have a low PMI.
- Just empirical probabilities — **no distributional model**, completely transparent.
- Weaknesses: ignores **estimation uncertainty**; discards **frequency**; **sensitive to rare words** (filter before use).

PMI is also used to grow dictionaries through **seed expansion** (more on this shortly). Jurafsky & Martin, ch. 6.

PMI on the Respondent Table

Take the five respondents from before (3 conservative, 2 liberal). The word *Jesus* appears in Respondents 1 and 2 — both conservative.

$$P(\text{Con}) = \frac{3}{5}, \quad P(\text{Jesus}) = \frac{2}{5}, \quad P(\text{Jesus}, \text{Con}) = \frac{2}{5}$$

$$\text{PMI}(\text{Jesus}, \text{Con}) = \log_2 \frac{2/5}{(2/5)(3/5)} = \log_2 \frac{5}{3} \approx 0.74 \quad (> 0)$$

- *Jesus* is **over-represented** among conservatives (PMI > 0).
- *family* appears in *every* survey: $\text{PMI}(\text{family}, \text{Con}) = \log_2 1 = 0$ — carries no group information.
- *Jesus* never co-occurs with *Lib* here $\Rightarrow \text{PMI} \rightarrow -\infty$: the rare-cell instability that makes us **filter or shrink before trusting PMI**.

Method 2 — Gentzkow & Shapiro (2010): the setup

Question: which phrases distinguish Republican from Democratic speech — and by how much?

Fix a phrase p (a two- or three-word collocation). Classify every token by **which party** produced it and **whether it is phrase** p . This gives a 2×2 **contingency table** of counts:

	uses p	not p	total
Republican	$O_{R,in}$	$O_{R,out}$	n_R
Democrat	$O_{D,in}$	$O_{D,out}$	n_D
total	f_p	$N - f_p$	N

O_{ga} : **observed** count in cell (party g , phrase status a). n_g : tokens by party g .
 f_p : total uses of p . N : all tokens. The four cells are constrained by their margins $(n_R, n_D, f_p, N - f_p)$.

Gentzkow & Shapiro: the χ^2 score

Null model: phrase use is **independent of party**. Then each cell's **expected** count is the product of its margins over the total,

$$E_{ga} = \frac{(\text{row total})_g (\text{column total})_a}{N} \quad \left(\text{e.g. } E_{R,\text{in}} = \frac{n_R f_p}{N} \right),$$

and we compare observed to expected across **all four cells**:

$$\chi_p^2 = \sum_{g \in \{D,R\}} \sum_{a \in \{\text{in,out}\}} \frac{(O_{ga} - E_{ga})^2}{E_{ga}}.$$

- Under the null each $(O_{ga} - E_{ga}) / \sqrt{E_{ga}} \approx N(0, 1)$; the fixed margins leave **one** free term, so $\chi_p^2 \sim \chi_1^2$ — a calibrated **noise benchmark**.
- **What this adds to PMI:** a phrase scores high only if it is over-represented *beyond* the sampling variation its frequency implies.
- G&S **rank** phrases by χ_p^2 and keep the top ones (not a formal test). The normal approximation needs non-tiny cells — the rare-phrase caveat returns.

G&S Stage 1: The Most Democratic Phrases

Two-Word Phrases

private accounts
trade agreement
American people
tax breaks
trade deficit
oil companies
credit card
nuclear option
war in Iraq
middle class

Three-Word Phrases

veterans health care
congressional black caucus
VA health care
billion in tax cuts
credit card companies
security trust fund
social security trust
privatize social security
American free trade
central American free

Phrases with the highest χ^2 association with *Democratic* speech in the
Congressional Record.

G&S Stage 1: The Most Republican Phrases

Two-Word Phrases

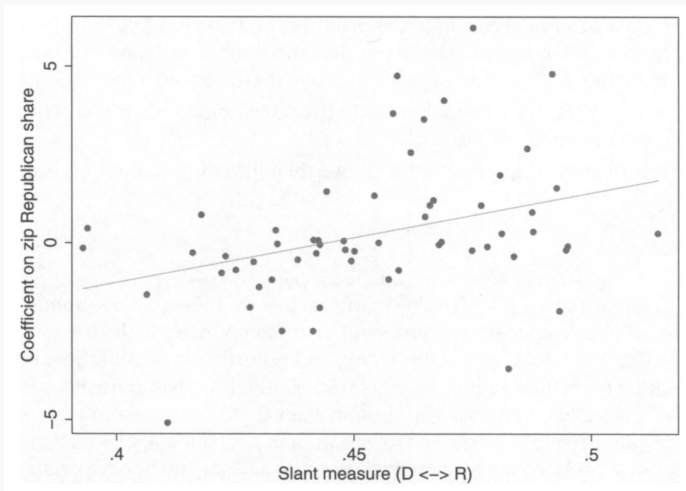
stem cell
natural gas
death tax
illegal aliens
class action
war on terror
embryonic stem
tax relief
illegal immigration
date the time

Three-Word Phrases

embryonic stem cell
hate crimes legislation
adult stem cells
oil for food program
personal retirement accounts
energy and natural resources
global war on terror
hate crimes law
change hearts and minds
global war on terrorism

The Republican counterpart. Note the framing pairs: *death tax* (R) vs. *estate tax* (D).

G&S Stage 2: Does the Slant Measure Track Reality?



Validation: newspapers in more-Republican markets (higher local Republican vote share) adopt more Republican slant. The measure tracks

Before Fightin' Words: Multinomial Counts

For each group g (Democratic vs. Republican, author A vs. author B), imagine a word-probability vector:

$$\boldsymbol{\mu}_g = (\mu_{g1}, \dots, \mu_{gJ}), \quad \sum_{j=1}^J \mu_{gj} = 1$$

The observed word counts are treated as draws from that group vocabulary:

$$\mathbf{W}_g = (W_{g1}, \dots, W_{gJ}) \sim \text{Multinomial}(n_g, \boldsymbol{\mu}_g)$$

- n_g is the total number of tokens in group g ; W_{gj} is the count of word j .
- The assumption: a group has stable word probabilities, and observed counts fluctuate around them.
- The payoff: rare words are treated as noisy estimates of μ_{gj} , not as decisive evidence by themselves.

From Multinomial to a Smoothed Log-Odds

What we want. Is word w more characteristic of group 1 or 2? Collapse the multinomial to “ w vs. everything else” and take the **log-odds ratio**:

$$\delta_w = \log \frac{\mu_w^{(1)}}{1 - \mu_w^{(1)}} - \log \frac{\mu_w^{(2)}}{1 - \mu_w^{(2)}}.$$

Why not just plug in $\hat{\mu}_w = y_w/n$? The same rare-cell problem as PMI: $y_w = 0$ gives $\log 0 = -\infty$, and for rare words $\hat{\mu}_w$ is very noisy, so δ_w swings wildly.

Fix: a Dirichlet prior (pseudocounts). Add α_w “imagined” prior occurrences to each word. Conjugacy gives the posterior-mean estimate

$$\hat{\mu}_w = \frac{y_w + \alpha_w}{n + \alpha_o}, \quad \alpha_o = \sum_w \alpha_w \text{ (total prior mass),}$$

so the “not- w ” count becomes $n + \alpha_o - y_w - \alpha_w$. Substitute into $\delta_w \Rightarrow$ the Fightin’ Words formula (next slide).

The Dirichlet Prior: Two Objects, Two Jobs

The prior looks like “many α ’s,” but it is **two objects**, each with a distinct job:

- α_w — pseudocount for **word** w (one per vocabulary word). **Job: shape** — which words the prior favors.
- $\alpha_o = \sum_w \alpha_w$ — the **total prior mass**. **Job: strength** — how many “pretend” tokens we inject.

Strength = data vs. prior. The smoothed estimate is a weighted average:

$$\hat{\mu}_w = \frac{y_w + \alpha_w}{n + \alpha_o} = \underbrace{\frac{n}{n + \alpha_o}}_{\text{weight on data}} \cdot \frac{y_w}{n} + \underbrace{\frac{\alpha_o}{n + \alpha_o}}_{\text{weight on prior}} \cdot \frac{\alpha_w}{\alpha_o}.$$

Large α_o (relative to n) shrinks hard; small α_o barely smooths.

Why a closed form? Conjugacy. Dirichlet prior + multinomial data $\Rightarrow$ Dirichlet posterior, so updating just **adds counts** ($\alpha_w \rightarrow \alpha_w + y_w$). No simulation — the estimate stays one line.

Choosing the α_w

The data fixes y_w, n ; **you** choose the prior. Two standard options:

- **Symmetric** (uninformative): $\alpha_w = \alpha$ for all w — ordinary Laplace / add- α smoothing, no word favored.
- **Informative** (Monroe et al. 2008): $\alpha_w \propto$ the word's frequency in the **pooled** corpus — common words get more prior mass, which tames the long tail and is the version that behaves best.

Reading the formula. α_w and α_o appear **four times** only because there are **two groups**, each contributing a numerator (word w) and a denominator (not- w):

$$\log \frac{y_w^{(g)} + \alpha_w}{\underbrace{n^{(g)} + \alpha_o - y_w^{(g)} - \alpha_w}_{\text{smoothed not-}w \text{ count}}}, \quad g \in \{1, 2\}.$$

Method 3 — Fightin' Words

Substituting the smoothed estimate $\hat{\mu}_w = (y_w + \alpha_w)/(n + \alpha_o)$ into the log-odds ratio δ_w gives the **Fightin' Words** statistic:

$$\hat{\delta}_w = \log \frac{y_w^{(1)} + \alpha_w}{n^{(1)} + \alpha_o - y_w^{(1)} - \alpha_w} - \log \frac{y_w^{(2)} + \alpha_w}{n^{(2)} + \alpha_o - y_w^{(2)} - \alpha_w}$$

Standardize by its uncertainty to get a z-score:

$$z_w = \frac{\hat{\delta}_w}{\sqrt{\widehat{\text{Var}}(\hat{\delta}_w)}}, \quad \widehat{\text{Var}}(\hat{\delta}_w) \approx \frac{1}{y_w^{(1)} + \alpha_w} + \frac{1}{y_w^{(2)} + \alpha_w}.$$

$y_w^{(i)}$: count of word w in group i ; $n^{(i)}$: total tokens in group i ; α_w : prior pseudocount for w ; $\alpha_o = \sum_w \alpha_w$.

- Each $\log(\cdot)$ is just the posterior log-odds of “w vs. not-w”; the denominator $n + \alpha_o - y_w - \alpha_w$ is the smoothed “not-w” count.
- The multinomial says what count variation to expect (the variance); the prior shrinks unstable rare-word differences toward zero — both for $\hat{\delta}_w$ and for its variance.

Fightin' Words: What You Get

The Dirichlet–multinomial model buys two things:

- **Shrinkage** — rare words are automatically pulled toward zero; no need for ad-hoc frequency filters
- **Interpretable z-score per term** — easy to rank, easy to plot, easy to communicate

This is the most principled of the three methods — but it also makes the most assumptions (a Dirichlet prior on word frequencies).

The Three Methods, Side by Side

	Uncertainty?	Shrinks rare?	Output	Assumes
PMI	no	no (must filter)	log-ratio	nothing (empirical probs)
G&S χ^2	yes (test)	no	rank / slant	independence null
Fightin' Words	yes (variance)	yes (prior)	z-score	Dirichlet–multinomial

- Start simple (PMI) to *see* the data; report a Fightin' Words z-score when you need a defensible ranking.
- The sequence adds structure for uncertainty: first a test benchmark, then prior-based shrinkage for rare terms.
- More assumptions can buy better behavior — *if* they hold. That is what **validation** checks.

Principles of Measurement

Measurement

Identify concepts relevant to our theory and create variables to quantify them:

- How can you link the concept to data?
- What information do you need to retain?

Principles of Measurement

- Measures should have **clear goals**
- The **source** of the original material should be identified
- The coding process should be **explainable and reproducible**
- The measure should be **validated**
- **Limitations** should be explored and discussed

Keyword Counting

Count instances of a particular word or phrase:

- + Easy to implement
- + Effective when specific keywords are strongly associated with the concept of interest
- + Clear and easy to communicate
 - o Often good to have, even if the main analysis is more complex
- Ineffective at capturing complex latent concepts
- Can detect whether a concept is mentioned, not necessarily *how* it is mentioned

Example: Economic Policy Uncertainty

Baker, Bloom & Davis (2016) build a keyword-counting index from newspapers. An article counts toward the index only if it contains a term from **all three** sets:

- **Economy:** economic, economy
- **Policy:** congress, legislation, regulation, deficit, Federal Reserve, ...
- **Uncertainty:** uncertain, uncertainty
- Scale each paper's count by its article volume, standardize, average across papers → a monthly **EPU index**.
- Transparent and cheap; spikes line up with elections, debt-ceiling fights, crises.
- ! **Validated** against a human audit of ~12,000 articles — the same habit we keep returning to.

Example: Duneier (2016)

Goal. Trace the **conceptual history** of ghetto — how its meaning migrated from Jewish enclaves (Venice, then Nazi Europe) to Black American neighborhoods.

Data. Google Ngrams (word frequencies in a books corpus). Each use of ghetto is split by context: “Warsaw”/“Jewish” vs. “black”/“Negro”.

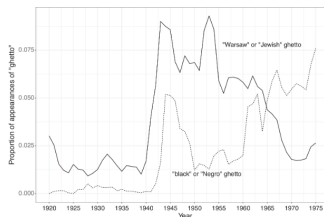


Figure 16.1. Graph 1 from Duneier (2016) depicting the proportion of the uses of the word ghetto in context of words signalling one of two uses. Data derived from Google Ngrams.

A **keyword count over time**, split by context, tracks a concept's **shifting meaning**: the Jewish sense leads before ~1950, the Black-American sense overtakes it by the late 1960s.

Dictionary Methods

From keywords to dictionaries:

- Generalization of keyword counting
- Based on lists of words or phrases
- Typically, weight frequency by the degree that each word is associated with the category of interest

$$\pi_i = \sum_{j=1}^J \frac{\mu_j W_{ij}}{M_i}$$

Where $M_i = \sum_{j=1}^J W_{ij}$ words are used in document i .

Dictionaries

- **Corpus-specific** keywords: tailored for the research question
 - E.g. patriotic terms; terms related to uncertainty; media slant
 - Made available by researchers as part of replication packages
 - Created using context-specific knowledge; human evaluators...
 - Important to validate the dictionary
- **General dictionaries**, some examples:
 - **Loughran–McDonald**: financial sentiment and uncertainty
 - LIWC: psychological categories (needs license)
 - Harvard Inquirer: general sentiment

Building a Dictionary: Esposito et al. (2023)

Esposito, Rotesi, Saia & Thoenig measure reconciliation vs. divisive narratives in the US press after the Civil War:

1. **Seed words** — candidate terms drawn from substantive sources (*Birth of a Nation* script, Lost Cause tenets, historical speeches). **External judges validate** their semantic relevance → 12 patriotic, 17 divisive seeds.
2. **PMI expansion** — on 300,000 pre-treatment *Chronicling America* pages (1904–1914), take each seed's top-1,000 PMI neighbors; retain words appearing across *multiple* seed lists.
3. **Re-validation** — judges inspect the augmented set and score semantic relevance; aggregate across judges → ranked bag of words.

Humans bracket the machine: validate the seeds; validate the expansion.

Sentiment Analysis

Objective: Determine if a document employs a *positive* or *negative* tone.

Traditionally done using dictionary methods:

- Obtain lists of positive and negative terms
- Count how many negative and positive words appear, possibly assign weights
 - **VADER** (tuned for social media)
 - **Loughran-McDonald** (tuned for finance)

This is the task many of you have in mind — and a natural first application of dictionary methods.

Sentiment Analysis: Key Considerations

- Sentiment scores may contain **biases** (the training corpus shapes what “positive” means)
- **Context is crucial:**
 - VADER assigns a negative score to cancer
 - What if the analysis pertains to the earnings reports of a pharmaceutical company?
- A sentiment score is a measurement like any other: **validate it against a ground truth before using it in inference**

Dictionaries and Validation

So: Validate

The habit the course is built around

Suppose you have built a sentiment measure, and you also happen to have a set of reviews with star ratings. One way to validate the measure:

1. Treat the **star ratings as a human-labeled benchmark** (e.g. 1 ★ = negative, 5 ★ = positive)
2. Compute your measure (dictionary / VADER score) on the same texts
3. Compare: **accuracy, confusion matrix**, rank correlation
4. **Read the misclassified cases** — where does the rule break? (negation, sarcasm, domain words)

The same logic works with any trusted label. In the Class 1 notebook we run it with song *genre* as the benchmark.

Validation Is Not Optional

The gap between measure and truth is **measurement error**.

- As an *outcome*: random error inflates standard errors; systematic error biases estimates.
- As a *regressor*: a generated-variable problem (Class 5).

A validation sample is exactly what lets you reason about — and later correct — this error.

Takeaways

1. A corpus is a **sample**; selection and representation are modeling choices — always justify them.
2. Preprocessing is modeling; the document-feature matrix and tf-idf are the foundation.
3. Discriminating words trade distinctiveness vs. prevalence: **PMI** → **Gentzkow-Shapiro** → **Fightin' Words**, by increasing assumptions.
4. Build dictionaries from seeds; **humans approve them** (Esposito et al.).
5. Any text-derived number is a measurement: **always validate**.

Live: Class 1 notebook. **Practice:** Problem Set 1.